

Estimates on the domain of validity for Lyapunov-Schmidt reduction

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Notations and Definitions

- $\mathbf{f} \in \mathcal{C}^\nu (\nu \in \mathbb{N}) \equiv \nu$ -times continuously differentiable map
- $D_{\mathbf{x}}\mathbf{f} \equiv$ Jacobian matrix $\left[\frac{\partial f_i}{\partial x_j} \right]$ of map $\mathbf{f}$ with respect to $\mathbf{x}$.
- $\mathcal{B}(\mathbf{x}_0, r) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_0\| < r\} \equiv$ open ball of radius $r > 0$ centred at $\mathbf{x}_0 \in \mathbb{R}^n$ where $\|\cdot\|$ is a vector norm on $\mathbb{R}^n$.
- $\|A\| := \sup\{\|A\mathbf{x}\| : \|\mathbf{x}\| = 1\} \equiv$ matrix norm induced by vector norm.
- $\mathcal{N}(\mathbf{x}) \equiv$ an open neighbourhood of $\mathbf{x}$.
- $\tanh(\mathbf{x}) := \left[\tanh(x_1) \quad \dots \quad \tanh(x_n) \right]^\top$ for $\mathbf{x} \in \mathbb{R}^n$.

Introduction

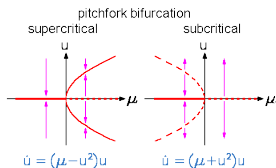


Figure: pitchfork bifurcation in 2 – dim system occurs along lower 1 – dim manifold; nature and number of equilibria change across $\mu = 0$

- **Bifurcation:** point in a parametrised non-linear dynamical system across which it undergoes **qualitative changes in dynamic behaviour** [GH13]¹.
- **Lyapunov-Schmidt reduction:** **dimensionality reduction** technique in non-linear systems analysis, used in the study of **bifurcation problems**.
- We present estimates for the region of validity of the reduction by applying bounds on the implicit function theorem derived in [JCB24]².

¹ Guckenheimer "Nonlinear oscillations, dynamical systems, and bifurcations of vector fields"

² Jindal et al. "Estimates of the size of the domain of the implicit function theorem: a mapping degree-based approach"

Implicit Function Theorem

Implicit Function Theorem — fundamental result in analysis.

Implicit Function Theorem (from [Zei93, Theorem 4.B]³)

Let $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ be open sets. Consider a $\mathcal{C}^\nu$ smooth map

$$U \times V \ni (\mathbf{x}, \mathbf{y}) \mapsto f(\mathbf{x}, \mathbf{y}) \in \mathbb{R}^m$$

where $\nu \geq 1$, and consider a point $(\mathbf{x}_0, \mathbf{y}_0) \in U \times V$ for which the map $D_{\mathbf{y}}f(\mathbf{x}_0, \mathbf{y}_0) : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is isomorphic. Then there exist neighbourhoods $\mathcal{N}(\mathbf{x}_0) \subset U, \mathcal{N}(\mathbf{y}_0) \subset V$ and unique smooth map $g : \mathcal{N}(\mathbf{x}_0) \rightarrow \mathcal{N}(\mathbf{y}_0)$ s.t.

$$f(\mathbf{x}, g(\mathbf{x})) = C = f(\mathbf{x}_0, \mathbf{y}_0) \quad \forall \mathbf{x} \in \mathcal{N}(\mathbf{x}_0)$$

Implication: existence of local unique smooth explicit representation $\mathbf{x} \xrightarrow{g} \mathbf{y}$ for the level set of f containing $(\mathbf{x}_0, \mathbf{y}_0)$ if nonsingular jacobian in $\mathbf{y}$

³Zeidler "Nonlinear functional analysis and its applications, Vol I: Fixed-point theorems" 

Explicit Bounds on the Implicit Function Theorem

In [JCB24, Corollary 3.8], explicit bounds on the domain $\mathcal{N}(\mathbf{x}_0)$ and co-domain $\mathcal{N}(\mathbf{y}_0)$ in which the implicit map hold were derived. Define:

- $M_x := \|D_x f(\mathbf{x}_0, \mathbf{y}_0)\|$ and $M_y := \|(D_y f(\mathbf{x}_0, \mathbf{y}_0))^{-1}\|$
- for $r_x, r_y > 0$ denote $\mathcal{B}_x := \mathbf{x} \in \mathcal{B}(\mathbf{x}_0, r_x)$ and $\mathcal{B}_y := \mathcal{B}(\mathbf{y}_0, r_y)$
- $\mathbf{L}_x(r_x) := \sup\{\|D_x f(\mathbf{x}, \mathbf{y}_0) - D_x f(\mathbf{x}_0, \mathbf{y}_0)\| : \mathbf{x} \in \mathcal{B}_x\}$
- $\mathbf{L}_y(r_x, r_y) := \sup\{\|D_y f(\mathbf{x}, \mathbf{y}) - D_y f(\mathbf{x}_0, \mathbf{y}_0)\| : (\mathbf{x}, \mathbf{y}) \in \mathcal{B}_x \times \mathcal{B}_y\}$

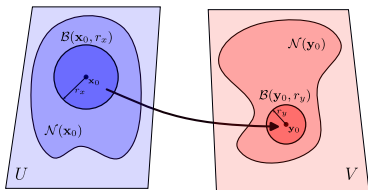


Figure: schematic of smooth implicit map $\varphi : \mathcal{B}_x \rightarrow \mathcal{B}_y$

Theorem on Bounds

for $r_x, r_y > 0$ simultaneously satisfying

$$\mathbf{L}_x(r_x)r_x + \mathbf{L}_y(r_x, r_y)r_y < \frac{r_y}{M_y} - M_x r_x$$

$$M_y \mathbf{L}_y(r_x, r_y) < 1$$

$\exists$ smooth map $g : \mathcal{B}_x \rightarrow \mathcal{B}_y$ that satisfies $f(\mathbf{x}, g(\mathbf{x})) = f(\mathbf{x}_0, \mathbf{y}_0)$

Lyapunov Schmidt reduction (introduction)

- Elementary bifurcations occur along a **center manifold**, most commonly 1–dimensional.
- **Goal:** obtain **reduced dimensional** representation of a high dimensional system around the singular (bifurcation) point.
- Local bifurcation diagram preserved after dimension reduction.
- The reduced order system is used to classify/analyse bifurcations

Lyapunov Schmidt reduction (algorithm)

- Smooth NLTI system $\dot{\mathbf{x}} = \Phi(\mathbf{x}, \lambda)$
- Singular point at $(\mathbf{x}_0, \lambda_0)$:
 - ▶ $\Phi(\mathbf{x}_0, \lambda_0) = 0$
 - ▶ $q := \dim(\ker(J)) > 0$
- Linear orthogonal projection operator $P : \mathbb{R}^n \rightarrow \text{range}(J)$

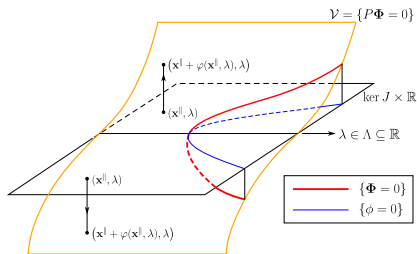
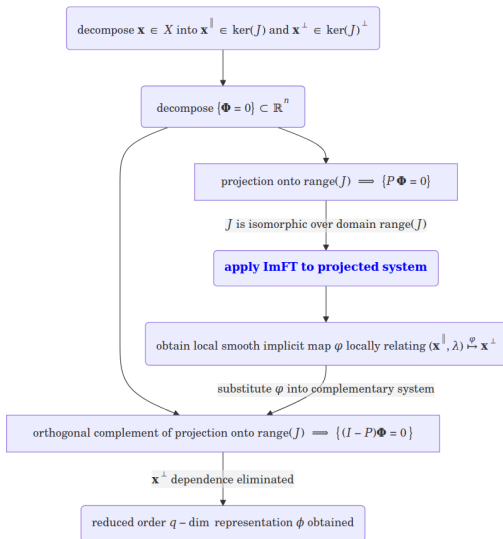
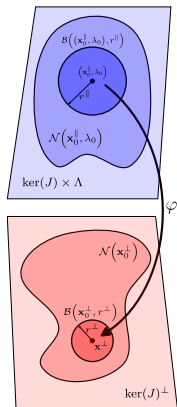


Figure: sketch of LSR for a 2 – dim system with a pitchfork bifurcation



Explicit Bounds on Lyapunov-Schmidt reduction



Bounds of validity for reduced dim map estimated at step where the Implicit Function Theorem is applied

Statement of Bounds

For any $r^{\parallel}, r^{\perp} > 0$ satisfying

$$\mathbf{L}_{(\mathbf{x}^{\parallel}, \lambda)}(r^{\parallel})r^{\parallel} + \mathbf{L}_{\mathbf{x}^{\perp}}(r^{\parallel}, r^{\perp})r^{\perp} < \frac{r^{\perp}}{M_{\mathbf{x}^{\perp}}} - M_{(\mathbf{x}^{\parallel}, \lambda)}r^{\parallel}$$

$$M_{\mathbf{x}^{\perp}}\mathbf{L}_{\mathbf{x}^{\perp}}(r^{\parallel}, r^{\perp}) < 1$$

$\exists$ a smooth implicit map $\varphi : \mathcal{B}^{\parallel} \rightarrow \mathcal{B}^{\perp}$ satisfying

$$\Phi(\mathbf{x}^{\parallel} + \varphi(\mathbf{x}^{\parallel}, \lambda), \lambda) = 0 \quad \forall (\mathbf{x}^{\parallel}, \lambda) \in \mathcal{B}^{\parallel}$$

$$M_{\mathbf{x}}, M_{\mathbf{y}}, r_{\mathbf{x}}, r_{\mathbf{y}}, \mathbf{L}_{\mathbf{x}}, \mathbf{L}_{\mathbf{y}} \rightarrow M_{(\mathbf{x}^{\parallel}, \lambda)}, M_{\mathbf{x}^{\perp}}, r^{\parallel}, r^{\perp}, \mathbf{L}_{(\mathbf{x}^{\parallel}, \lambda)}, \mathbf{L}_{\mathbf{x}^{\perp}}$$

are respective **translations of quantities** defined in slide [6] **into the Lyapunov-Schmidt reduction framework.**

Example: Recurrent Neural Network model

System: for $\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^T \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$, consider the dynamic model of the following two-node RNN:

$$\dot{\mathbf{x}} = -\mathbf{x} + \tanh \left(\begin{bmatrix} 0 & \lambda \\ \lambda & 0 \end{bmatrix} \mathbf{x} \right)$$

Final values of bound parameters:

- $M_{(\mathbf{x}^{\parallel}, \lambda)} = 0$, $L_{(\mathbf{x}^{\parallel}, \lambda)}(r^{\parallel}) = 0$, $M_{\mathbf{x}^{\perp}} = \frac{1}{2}$

$$L_{\mathbf{x}^{\perp}}(r^{\parallel}, r^{\perp}) = \{1 - \min(0, \lambda) :$$

- $$\left(\frac{1}{\sqrt{2}}(x_1 + x_2), \lambda \right) \in \mathcal{B}((0, 1), r^{\parallel})\}$$

$$= 1 - (1 - r^{\parallel}) = r^{\parallel}$$

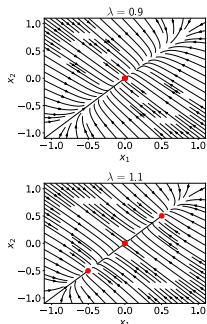


Figure: system flow around pitchfork bifurcation at origin near $\lambda_0 = 1$

Example: Recurrent Neural Network model

Substituting the parameters finally yields $r^\perp \in \mathbb{R}^+$ and $r^\parallel \in (0, 2)$

Results

Let $\mathbf{v} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \end{bmatrix}^\top$. For $r^\perp > 0, r^\parallel \in (0, 2)$ we have the implicit map $\varphi : \mathcal{B}((0, 1), r^\parallel) \rightarrow \mathcal{B}(0, r^\perp)$ satisfying

$$\varphi(\alpha \mathbf{v}, \lambda) + \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \tanh \left(\lambda (\alpha \mathbf{v} - \varphi(\alpha \mathbf{v}, \lambda)) \right) = 0$$

yielding reduced dim system $g(\alpha, \lambda) = -\alpha + \mathbf{v}^\top \tanh \left[\lambda (\alpha \mathbf{v} - \varphi(\alpha \mathbf{v}, \lambda)) \right]$

Example: RNN model (maximal bounding ball)

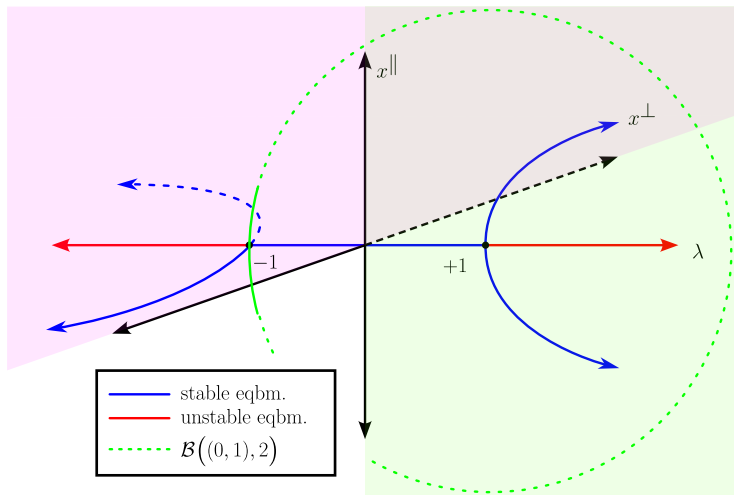


Figure: bifurcation diagram for illustrated 2-node RNN model

Extensions and Future Plans

- Application to higher dimensional systems including nonlinear opinion dynamics [BFL22]¹ to explore how the graph structure affects the tightness of the established bounds.
- Extension to infinite-dimensional systems, since the Lyapunov Schmidt reduction is defined more generally for Fredholm operators over Banach spaces.

¹Anastasia Bizyaeva, Alessio Franci, and Naomi Ehrich Leonard. “Nonlinear opinion dynamics with tunable sensitivity”. In: *IEEE Transactions on Automatic Control* 68.3 (2022), pp. 1415–1430.